

BO PENG

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PROFESSIONAL SUMMARY

Frontend Software Engineer with 5+ years of experience building and shipping production React and TypeScript applications at scale, including AI-powered data visualization features used by 10,000+ enterprise customers at Amazon AWS QuickSight. Strong background in component architecture, REST API integration, automated testing (Jest, Playwright), CI/CD, and performance optimization across modern HTML, CSS, and JavaScript stacks.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), HTML5, CSS3, Python, Java, SQL

Frontend: React, Redux, Redux-Saga, React Query, React Flow, Vue.js, Vite, styled-components, responsive design, accessibility (WCAG), localization / i18n

Testing: Playwright (end-to-end), Jest, Mocha, unit & integration testing

Backend & Cloud: Node.js, Express, GraphQL, REST APIs, AWS (Lambda, DynamoDB, CloudWatch, CDK, Bedrock), MongoDB

Tools & Practices: Git, CI/CD pipelines, Agile/Scrum, code review, on-call operations, LLM/generative AI integration

PROFESSIONAL EXPERIENCE

Frontend Engineer II — Amazon Web Services (AWS), QuickSight

Jan 2022 – Apr 2026

New York, NY

- Led a team of 5+ engineers to architect and ship QuickSight Scenarios, a node-based visual editor built with React, TypeScript, and React Flow, enabling 5,000+ QuickSight Pro enterprise customers to interactively explore data pipelines.
- Integrated Amazon Bedrock LLM capabilities into the Scenarios frontend, delivering natural-language data querying and generative AI question recommendations that increased engagement across thousands of active accounts.
- Built the New Reader Experience Topics feature, improving Q&A discoverability and reducing average time-to-insight for 10,000+ enterprise users.
- Developed reusable React preview components rendering AI-agent-generated Office and PDF artifacts for QuickSuite Research end users.
- Authored Playwright end-to-end test suites covering Topics, Scenarios, and Stories across all test and production regions, ensuring full regression coverage in CI/CD pipelines.
- Automated account provisioning and test-resource setup for new region rollouts with AWS CDK (infrastructure as code), eliminating 5+ days of manual effort per region and accelerating global launches.
- Owned on-call operations for 10+ production UI and service pipelines, maintaining consistent uptime and monitoring via CloudWatch across all regions.

Software Engineer — Fanbyte

Feb 2021 – Jan 2022

New York, NY

- Developed modular npm microservices for the advertising platform and integrated them into the Vue.js front end, improving ad delivery and web-traffic capture.
- Built RESTful API endpoints in Node.js/Express surfacing publisher revenue analytics in a redesigned Vue.js dashboard, enabling real-time reporting for internal stakeholders.
- Wrote Mocha integration tests to validate microservice behavior, and rebuilt the company website with a custom WordPress theme using Vue components.

Software Developer — ManageGo

Aug 2019 – Dec 2019

Brooklyn, NY

- Implemented new responsive website designs in React with styled-components, improving user experience for thousands of property-management users.
- Created custom React hooks for query-string handling and managed application state with Redux and Redux-Saga; designed the API schema for a new concierge-facing commercial feature.
- Wrote unit tests for Redux actions, reducers, and sagas to ensure reliable application state.

EDUCATION

New York University — B.A. in Computer Science, Minor in Economics & Web Design

Sep 2014 – May 2018